🧊 M003 Extension — English Worksheet (Advanced)

Topic: FizzBuzz Pixel Art (remainders & rules) · **Course:** 3D Coordinates · **Level:** Extension (Advanced, after Week 3) · **Time:** about 45 minutes

You already made pixel art by choosing each block's colour **by hand**. This time the **code** chooses every colour for you — using one rule, the **FizzBuzz rule**. Load the world, type `run` in the chat, and watch the agent paint a **15 × 15** wall one block at a time, counting from **1 to 225**.

If the block's number...	it says	colour
divides by **3 and 5**	**FIZZ BUZZ**	🟡 gold
divides by **3**	**FIZZ**	⬛ black
divides by **5**	**BUZZ**	🟪 purple
divides by neither	*(nothing)*	🟩 green

> 🔢 **Big idea:** a block's colour comes from its **number**, not its place. The code asks "*what is the remainder when I divide this number by 3? by 5?*" — and the remainder picks the colour.

Remainder is what is left over after you divide. 6 ÷ 3 = 2 with **remainder 0** — 3 fits exactly. 7 ÷ 3 = 2 with **remainder 1** — one is left over. When the remainder is **0**, the number **divides evenly**.

1 · Predict 🔮

The wall counts **1** → **225**. Using the rule, write the **colour of each block and the reason**.

Block 9 →

Block 10 →

Block 15 →

Block 7 →

The rule checks **"divides by 3 and 5"** *first*, before it checks "divides by 3".

What colour would block 15 get if the code checked "divides by 3" first instead? Why does the order of the checks change the answer?

2 · Spot the Bug 🐛

Each rule below is broken. Read what it should do, fix it, then explain why the original was wrong and your fix works.

Bug A — the order is upside down. This should paint gold when a number divides by 3 **and** 5. But the **÷ 3** check comes first, so it grabs those numbers before the gold check ever runs.

```
if remainder of N ÷ 3 = 0:
    paint BLACK    ← FIZZ
else if remainder of N ÷ 3 = 0 and remainder of N ÷ 5 = 0:
    paint GOLD     ← FIZZ BUZZ
else if remainder of N ÷ 5 = 0:
    paint PURPLE   ← BUZZ
else:
    paint GREEN
```

Which blocks (15, 30, 45 ...) come out the wrong colour, and what colour do they wrongly get?

Write the fixed rule (which line moves where?):

Bug B — wrong divisor. The purple **BUZZ** stripe should mark blocks that divide by **5**. This line divides by **4**, so the wrong blocks turn purple.

```
else if remainder of N ÷ 4 = 0:
    paint PURPLE   ← BUZZ
```

Write the fixed line, then name two blocks that turned purple by mistake:

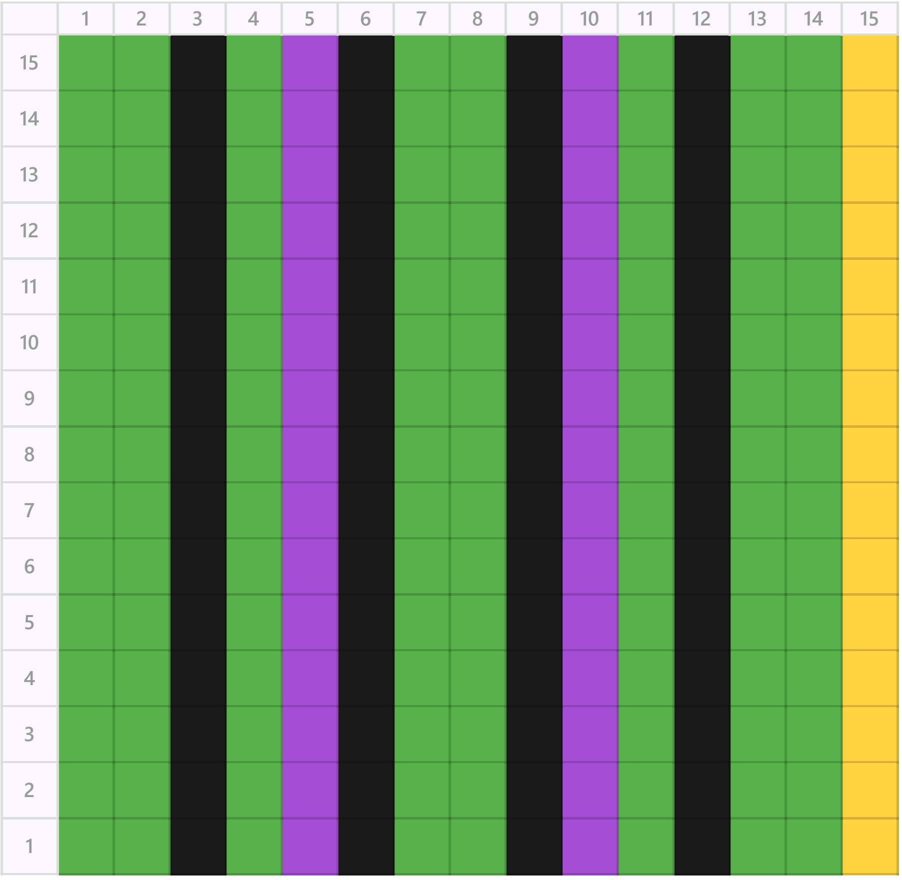
Bug C — wrong wall width. A friend changed the inner loop from **repeat 15** to **repeat 10**, so each row is only **10** blocks wide. The neat vertical stripes turned into a slanted, messy pattern.

Why do the stripes only line up when each row is 15 wide? (Hint: 3 × 5 = 15. What happens to the count at the start of every new row?)

3 · Show Your Work 📸 🎥

Load the world. Stand where you can see the empty wall. Type `run` in the chat and watch the agent count **1** → **225**, painting each block by the rule.

🧱 The finished wall — 15 × 15 (block 1 → 225)



Look closely. Every one of the 15 rows is **identical**, so the colours form **vertical stripes**.

Explain why every row comes out the same. (Each row is 15 wide, and 15 divides by both 3 and 5 — so what number does the *first block of every new row* act like?)

Modify challenge. Change **one rule** and **predict the new wall before you run it**. For example: paint gold for **÷ 2 and ÷ 3**, or make the wall **20** wide. Write which stripes will move, then build it and check.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the colours and the stripes.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.
>
> The most fun part was _____.
>
> Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.